

COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

FYP-I PROJECT PROPOSAL

IntelliSafe

Real-Time Working Site Safety Monitoring using Edge Computer Vision

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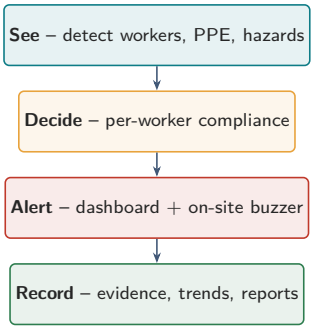
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Abstract

IntelliSafe is an edge-based safety monitoring system for industrial and construction sites. A camera on an **NVIDIA Jetson Nano** runs a single **YOLOv8n** detector for *worker, helmet, vest, fire* and *smoke*. A **compliance engine** links each helmet and vest to its worker, keeps a rolling compliance score, and raises *one* alert per continuing violation with an image as evidence.

Events travel over **MQTT** to a backend that stores them, streams them to a supervisor dashboard and produces PDF compliance reports. For fire or smoke, an **ESP32 alarm unit** sounds a buzzer on site (target: within one second), even if the internet link is down.



Motivation & Problem Statement

Why it matters

- ILO: ≈3 million work-related deaths and 395 million non-fatal injuries per year [1].
- Construction and manufacturing carry a large share; missing PPE and unnoticed fire or smoke are recurring causes.

Why current practice fails

- **Coverage gaps** – periodic rounds miss violations in between.
- **Human limits** – one supervisor cannot watch every worker all shift.
- **No evidence** – incidents are rebuilt from memory.
- **No trends** – nobody knows which zone or shift is the weak point.

Problem statement

Design and implement a **low-cost edge computer-vision platform** that monitors a work site in real time, detects PPE non-compliance and fire or smoke, and delivers **alerts, on-site alarms, image evidence** and **compliance analytics** to supervisors.

Approved detection scope: worker, helmet, vest, fire, smoke.

Literature Review – Prior Work

Work	Method	Data / Platform	Reported result	Open issue for a real site
Nath et al. [2] Autom. Constr. 2020	YOLOv3, three approaches; best one detects a worker and classifies PPE state (none / hat / vest / both)	Pictor-v3, ∼1.5k images; laptop	**72.3% mAP** at 11 FPS	Image-level verdict; no hazards, no alerting, evidence or trends
Wang et al. [3] Sensors 2021	Eight YOLO v3/v4/v5 detectors compared	CHV, 1,330 images, 6 classes; Tesla P40 GPU, PC / cloud	YOLOv5x **86.55% mAP**; YOLOv5s **52 FPS**	Detects PPE items only; no worker-level decision; server GPU
de Venâncio et al. [7] Neural Comput. Appl. 2022	YOLOv4 fire/smoke detector + filter pruning	D-Fire, 21k+ images; Raspberry Pi 4	Up to **83.6%** less compute, **83.9%** less memory, no accuracy loss	Fire only; no PPE, no alert delivery or dashboard

Detector basis: YOLO single-pass detection [4] and its current release, Ultralytics YOLOv8 [5], whose nano variant we deploy.

Takeaway. Detection of PPE [2, 3] and fire [7] is solved well enough to build on. What these works leave open is the *system around the detector*.

Research Gap & Our Contribution – How IntelliSafe Differs

Capability	[2]	[3]	[7]	IntelliSafe
Helmet + vest detection	✓	✓	–	✓
Fire / smoke detection	–	–	✓	✓
PPE + hazards in *one* model	–	–	–	✓
Worker-level compliance verdict	✓	–	–	✓
Temporal debounce, priorities	–	–	–	✓
Embedded / edge inference	–	–	✓	✓
Event delivery (MQTT)	–	–	–	✓
Image evidence + history	–	–	–	✓
Dashboard, trends, PDF reports	–	–	–	✓
On-site audible alarm (ESP32)	–	–	–	✓

✓ addressed in the paper – not reported in the paper

Contributions of this FYP

1. **Unified 5-class edge detector** for PPE and hazards on a Jetson Nano.
2. **Compliance engine**: IoU matching, rolling score, *N*-frame debounce and priority levels.
3. **End-to-end event pipeline**: MQTT, storage, evidence, live dashboard and reports.
4. **Local emergency alarm**: ESP32 buzzer for fire and smoke.
5. **On-device evaluation**: mAP, FPS, latency and false alerts per hour.

We do *not* claim a new detector. The contribution is a deployable, measured system.

Objectives & Research Questions

Objectives

1. **Detect** worker, helmet, vest, fire and smoke with YOLOv8n.
2. **Decide** per-worker compliance, one alert per violation, with evidence.
3. **Deploy** detector and engine in real time on the Jetson Nano.
4. **Deliver** events over MQTT; sound an **ESP32 buzzer** on fire or smoke.
5. **Store** events, compliance history and evidence images.
6. **Present** live alerts, trends and PDF reports on a dashboard.

Research questions

RQ1. Can one YOLOv8n model on a Jetson Nano detect all five classes in real time with mAP@0.5 ≥ 0.70?

RQ2. How much does N-frame debouncing cut false alerts per hour compared with frame-level alerting?

RQ3. What end-to-end latency is achieved from camera to dashboard and from camera to buzzer?

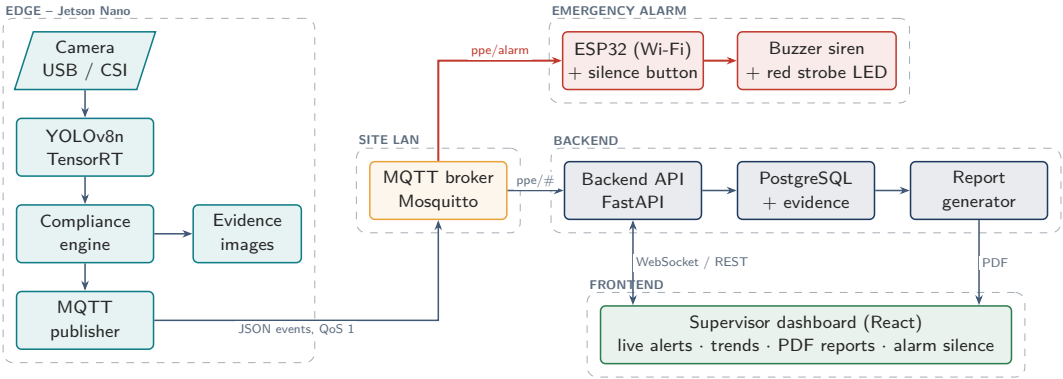
Deliverables & SDG Mapping

Deliverables at the end of FYP

1. Annotated dataset and trained YOLOv8n model (5 classes).
2. Jetson Nano application: detection, compliance engine, evidence.
3. ESP32 alarm unit: buzzer and strobe, driven over MQTT.
4. Backend with database for events, history and evidence.
5. Web dashboard with live alerts, trends and PDF reports.
6. FYP report, documentation and a live end-to-end demo.

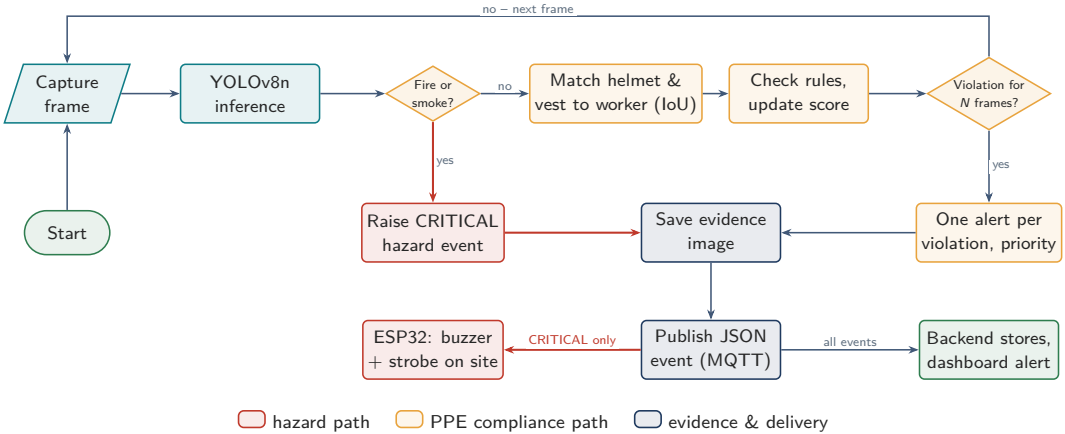
SDG	How IntelliSafe contributes
SDG 3 Good Health	Early PPE and fire/smoke detection prevents injuries.
SDG 8 Decent Work	Continuous monitoring supports safe, decent work.
SDG 9 Innovation	Low-cost edge AI brings evidence-based safety to industry.

Methodology – System Architecture



The broker runs on the site LAN, so the buzzer still sounds if the backend or internet link is down. Event format is fixed up front, so model and platform work proceed in parallel; a simulator feeds the backend until the Jetson is ready.

Methodology – Flow Chart



The edge loop runs every frame; the backend and alarm unit react asynchronously.

Methodology – Detection Model & Dataset

One model, three detection tracks

Track	Class	Used for
Worker	worker	Anchor for the PPE check, tracking
PPE	helmet	Violation: `helmet_missing`
PPE	vest	Violation: `vest_missing`
Hazard	fire	CRITICAL event + buzzer
Hazard	smoke	CRITICAL event + buzzer

Model & training

- YOLOv8n [4, 5]: nano variant, fits the Jetson Nano [6].
- 640×640 input, fine-tuned from COCO weights.
- Export `.pt` → ONNX → TensorRT (FP16).

Dataset

- Public sets from the literature: Pictor-v3 [2], CHV [3] for PPE, D-Fire [7] for fire and smoke.
- Plus self-collected local frames; YOLO format; train / val / test split.

Methodology – Compliance Engine

1. PPE–worker matching. A helmet or vest belongs to the worker whose box it overlaps most (IoU > 0.3); a simple tracker keeps a stable `worker_id`.

2. Rule. Helmet *and* vest required for every worker; fire or smoke skips the worker check.

3. Rolling score.

$$\text{Score} = \frac{\sum \text{compliant worker-frames}}{\sum \text{worker-frames}} \times 100$$

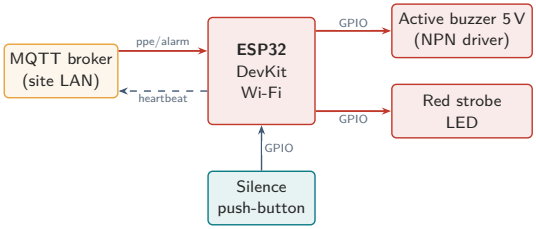
4. Debounce. Alert only after N consecutive violating frames, then once per worker per violation type.

5. Priority & action

Event	Priority	Action
Fire / smoke	CRITICAL	ESP32 buzzer + dashboard
Helmet *and* vest missing	HIGH	Dashboard alert
Helmet *or* vest missing	MEDIUM	Alert + log
Low compliance score	LOW	Flag in report

6. Evidence. Every event saves the annotated frame and a record: time, worker, class, confidence, image path.

Methodology – Emergency Alarm Unit (ESP32 + Buzzer)



Alarm message on `ppe/alarm`:

```
{"cmd":"on", "reason":"fire", "zone":"Z1"}
```

Behaviour

- Subscribes to `ppe/alarm` (QoS 1); fire or smoke turns on the siren and strobe.
- Only CRITICAL events sound the buzzer, so workers do not learn to ignore it.
- Silenced by the push-button or from the dashboard; turns off once the hazard clears.

Reliability

- Heartbeat every 5 s plus an MQTT *Last Will* [8]; the dashboard shows "alarm offline" when it stops.
- Works without internet, because the broker is on the site LAN.

Why ESP32

- Low cost, built-in Wi-Fi, mature MQTT libraries; the Jetson stays free for inference.

Methodology – Event Pipeline & Technology Stack

MQTT topics [8]

Topic	From → to	Content
`ppe/alerts`	Jetson → backend	violations, hazards, evidence path
`ppe/metrics`	Jetson → backend	rolling compliance score
`ppe/heartbeat`	Jetson, ESP32 → backend	device health
`ppe/alarm`	Jetson, dashboard → ESP32	siren on / off

Every event is JSON: type, event, priority, timestamp, worker_id, confidence, compliance_score, detections, image_path. QoS 1, buffered on the Jetson if the network drops.

Technology stack

Edge	Jetson Nano, JetPack, Python, Ultralytics YOLOv8, TensorRT, OpenCV, paho-mqtt
Alarm	ESP32, Arduino framework, PubSubClient, active buzzer
Messaging	Eclipse Mosquitto (MQTT)
Backend	FastAPI, PostgreSQL, WebSockets, ReportLab, pytest
Frontend	React + Vite, Recharts, Tailwind CSS
Tooling	Roboflow / CVAT, PyTorch, Colab, Git, Docker

Evaluation Plan

Metric	Target	Method
mAP@0.5 (test set) RQ1	≥ 0.70	Ultralytics val
Per-class precision / recall	reported	Confusion matrix
Inference speed on Jetson RQ1	real time	On-device benchmark
False alerts per hour RQ2	debounce vs. none	Live session
Camera → dashboard delay RQ3	measured	Timestamps
Camera → buzzer delay RQ3	$\leq 1\,\text{s}$	Timestamps

Test scenarios

- Compliant worker → green box, no alert.
- Helmet removed → one alert, evidence saved.
- Two workers, one non-compliant → per-worker result.
- Fire/smoke video → CRITICAL alert and **buzzer sounds**.
- ESP32 unplugged → "alarm offline" on the dashboard.
- Network drop → events buffered, delivered on reconnect.

Work Distribution

Uzair – Perception

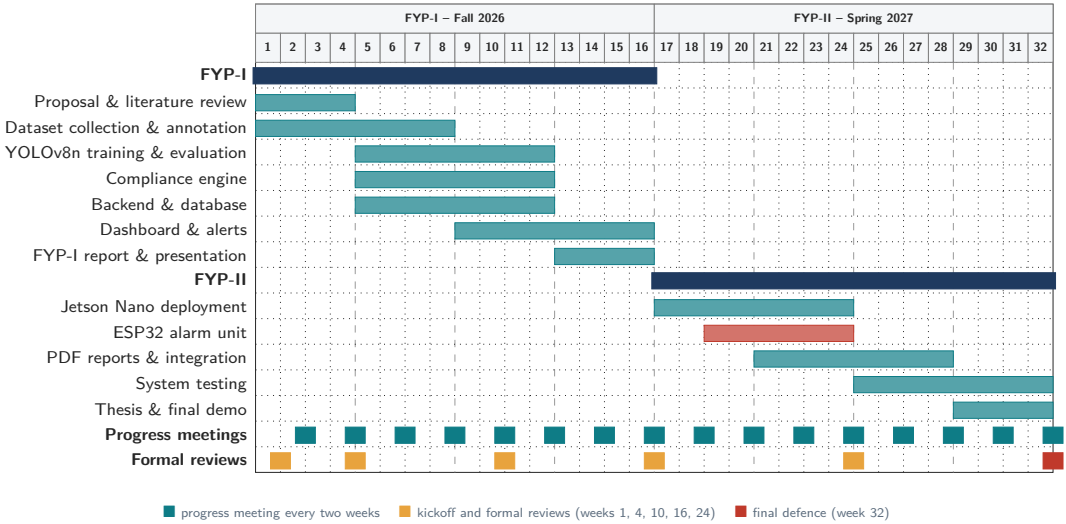
- Dataset collection and annotation
- YOLOv8n training, evaluation, export
- Compliance engine: matching, score, debounce
- Evidence capture and the event handoff

Muhammad Ahmad – Platform

- Jetson Nano setup and model deployment
- MQTT broker, publisher and backend
- ESP32 alarm unit: firmware and wiring
- Dashboard and PDF reports

Joint: event format · integration · testing · final demo and defence

Timeline – Gantt Chart (Two Semesters)



Timeline – Milestones & Supervisor Meetings

Week	Meeting	Evidence shown to the supervisor
1	Kickoff	Scope agreed: worker, helmet, vest, fire, smoke
4	Proposal review	Proposal approved; dataset sources and tools confirmed
10	Mid-term (FYP-I)	First trained model; compliance engine running on a laptop
16	FYP-I evaluation	Laptop demo from camera to dashboard; FYP-I report
24	Mid-term (FYP-II)	Model on the Jetson Nano; live events stored; ESP32 buzzer working
32	Final defence	Full live demo, PDF reports, thesis and demo video
Every 2 weeks	Progress meeting	Work done, problems, plan for the next two weeks

Conclusion

- PPE detection [2, 3] and edge fire detection [7] are proven; real sites still lack the **system around the detector**.
- IntelliSafe adds that system: a unified edge detector, per-worker compliance with debounce, an MQTT event pipeline, evidence, analytics, and an **on-site ESP32 alarm**.
- Success is measured on the device: mAP, FPS, false alerts per hour, and latency to the dashboard and to the buzzer.

Expected impact

From periodic inspection to **continuous, evidence-based** monitoring: violations are caught as they happen, fire is announced on site within a second, and managers see the trend before an accident does.

References

[1] International Labour Organization, *A Call for Safer and Healthier Working Environments*, Geneva, 2023.

[2] N. D. Nath, A. H. Behzadan, and S. G. Paal, "Deep learning for site safety: Real-time detection of personal protective equipment," *Automation in Construction*, vol. 112, 103085, 2020.

[3] Z. Wang, Y. Wu, L. Yang, A. Thirunavukarasu, C. Evison, and Y. Zhao, "Fast personal protective equipment detection for real construction sites using deep learning approaches," *Sensors*, vol. 21, no. 10, 3478, 2021.

[4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, real-time object detection," in *Proc. IEEE CVPR*, 2016, pp. 779–788.

[5] G. Jocher, A. Chaurasia, and J. Qiu, *Ultralytics YOLOv8*, 2023. `https://github.com/ultralytics/ultralytics`

[6] NVIDIA Corporation, *Jetson Nano Developer Kit*. `https://developer.nvidia.com/embedded/jetson-nano`

[7] P. V. A. B. de Venâncio, A. C. Lisboa, and A. V. Barbosa, "An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices," *Neural Computing and Applications*, vol. 34, pp. 15349–15368, 2022.

[8] OASIS, *MQTT Version 5.0*, OASIS Standard, March 2019. `https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.html`

Thank You

Questions & Discussion

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